

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

NumPy Library : Functions

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NumPy

NumPy (Numerical Python) is an open source Python library that's widely used in science and engineering. The NumPy library contains multidimensional array data structures, such as the homogeneous, N-dimensional ndarray, and a large library of functions that operate efficiently on these data structures.

Table 1: Commonly Used NumPy Functions and Their Descriptions

Function	Description
<code>numpy.array()</code>	Creates a NumPy array from a list or tuple.
<code>numpy.zeros()</code>	Creates an array filled with zeros.
<code>numpy.ones()</code>	Creates an array filled with ones.
<code>numpy.arange()</code>	Returns evenly spaced values within a specified interval.
<code>numpy.linspace()</code>	Returns evenly spaced numbers over a specified range.
<code>numpy.reshape()</code>	Changes the shape of an array without changing its data.
<code>numpy.ravel()</code>	Flattens a multi-dimensional array into one dimension.
<code>numpy.sum()</code>	Computes the sum of array elements.
<code>numpy.mean()</code>	Computes the arithmetic mean of array elements.
<code>numpy.std()</code>	Computes the standard deviation of array elements.
<code>numpy.min()</code>	Returns the minimum value in an array.
<code>numpy.max()</code>	Returns the maximum value in an array.
<code>numpy.argmin()</code>	Returns the index of the minimum value.
<code>numpy.argmax()</code>	Returns the index of the maximum value.
<code>numpy.unique()</code>	Finds unique elements in an array.
<code>numpy.sort()</code>	Sorts the elements of an array.
<code>numpy.where()</code>	Returns indices of elements that satisfy a condition.
<code>numpy.concatenate()</code>	Joins arrays along an existing axis.
<code>numpy.vstack()</code>	Stacks arrays vertically.
<code>numpy.hstack()</code>	Stacks arrays horizontally.
<code>numpy.dot()</code>	Computes the dot product of two arrays.
<code>numpy.matmul()</code>	Performs matrix multiplication.
<code>numpy.linalg.inv()</code>	Computes the inverse of a matrix.

Function	Description
numpy.linalg.det()	Computes the determinant of a matrix.
numpy.save()	Saves an array to a binary file.
numpy.load()	Loads an array from a binary file.

For details on operations in pandas, see the official documentation: https://numpy.org/doc/stable/user/absolute_beginners.html